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ITAI 1371 – Machine Learning

Final Machine Learning Project

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Reflection Journal

Completing this machine-learning project helped me better understand the full process of building and evaluating a classification model. Before training the models, I learned that the dataset had to be carefully inspected and cleaned. One important issue was that the TotalCharges column appeared to have no missing values, but it actually contained 11 blank entries. Converting the column to numerical values showed why checking data types and hidden values is important.

I also learned how feature engineering can improve a dataset. I created TenureGroup to organize customers by how long they had remained with the company and NumberOfServices to count the services used by each customer. These features helped summarize customer behavior in a way that the models could understand.

Another important lesson was the purpose of training, validation, and test sets. The training set was used to teach the models, the validation set was used to compare their performance, and the test set was kept separate until the final evaluation. This helped prevent the test results from influencing model selection. I also learned that preprocessing must be fitted only on the training data to prevent data leakage.

The target variable was imbalanced because there were more customers who did not churn than customers who churned. I used RandomOverSampler only on the training set to give both classes equal representation. This taught me that balancing the validation and test sets would create unrealistic results because those sets should represent the original customer population.

The most interesting part of the project was comparing the six individual models and the two ensemble models. Gradient Boosting was the strongest individual validation model, while K-Nearest Neighbors produced the weakest results. The Bayesian Ensemble produced the highest final test ROC-AUC of 0.8531. It also achieved a recall of 0.7893, meaning that it identified approximately 79% of customers who actually churned.

This project also showed me that accuracy is not enough when evaluating an imbalanced classification problem. Precision, recall, F1-score, and ROC-AUC provided a more complete understanding of model performance. Recall was especially important because missing a customer who is likely to churn could prevent the company from taking retention action.

One challenge I experienced was keeping track of the different model predictions and probability variables. I also had to correct missing imports and make sure that every model was included in

the comparison tables. These challenges taught me the importance of running notebook cells in order, using clear variable names, and checking results before moving to the next step.

In the future, I would improve this project by using automated hyperparameter tuning, testing additional balancing methods such as SMOTE, and comparing different classification thresholds. I would also use explainability tools to show why individual customers received certain churn predictions.

Overall, this project improved my confidence in data cleaning, feature engineering, preprocessing, model training, ensemble learning, and evaluation. It gave me practical experience with the complete machine-learning workflow and helped me understand how machine learning can support customer-retention decisions.